

Mohamed Rayan Barhdadi

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Research Interests

1. Physics-informed and cognitive-science approaches to vision
2. 3D/4D Scene understanding and reconstruction
3. Vision-Language Models (VLMs) and video understanding

Education

Texas A&M University

2023 – 2027

Bachelor of Science, Electrical Engineering – (Qatar Campus)

- *Thesis (3rd year): “The Physical World as the First Teacher: Structured 4D Scene Representation.”*
Advisors: Prof. Hasan Kurban and Prof. Erchin Serpedin.
- *Notable Award:*
 - Richard E. Ewing Award for Excellence in Student Research
(1 out of all students; awarded to one undergraduate annually by faculty nomination).
- *Selected Summer/Winter Schools (typically for PhD students):*
 - International Computer Vision Summer School (ICVSS), Italy, 2026.
 - African Computer Vision Summer School (ACVSS), Ghana, 2026.
 - London Geometry and Machine Learning (LOGML), England, 2026.

Publications (Google Scholar)

* denotes equal contribution

IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video

R Khanbayov*, MR Barhdadi*, E Serpedin, and H Kurban

European Conference on Computer Vision

ECCV, 2026

EMPATHIA: Multi-Faceted Human-AI Collaboration for Refugee Integration

MR Barhdadi, M Tuncel, E Serpedin, and H Kurban

Neural Information Processing Systems (Main Proceedings, CreativeAI)

NeurIPS, 2025

+ Invited Keynote Talk at NeurIPS Affinity Workshop 2025

PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views

MR Barhdadi, H Kurban, and H Alnuweiri

Building Physically Plausible World Models Workshop

International Conference on Machine Learning

ICML–W, 2025

Under Review

4D Synchronized Fields: Motion-Language Gaussian Splatting

MR Barhdadi, S Abdaljalil, R Khanbayov, E Serpedin, and H Kurban

AAAI Conference on Artificial Intelligence

AAAI, 2027 (under rev.)

Intrinsic 4D Gaussian Segmentation from Scene Cues

H Yazar*, MR Barhdadi*, E Serpedin, M Tuncel, and H Kurban
British Machine Vision Conference

BMVC, 2026 (under rev.)

Perception, Not Reasoning, Limits Visual Theory of Mind

MR Barhdadi, ST Wasim, J Gall, E Serpedin, and H Kurban
Neural Information Processing Systems
 Also Initial Version Accepted at 2nd Workshop on MMRAgI

NeurIPS, 2026 (under rev.)
 CVPR-W, 2026

Mixture Normalizing Flows with Exact Likelihood

MR Barhdadi, SB Belhaouari, H Bensmail, and M Aupetit
Neural Information Processing Systems

NeurIPS, 2026 (under rev.)

Research Experience**Harvard University, AI and Robotics Lab**

03/2026 – Present

Research Assistant with Prof. Mengyu Wang and Dr. Minghan Li
 Topic: 3D/4D Understanding, Vision Language Models.

Texas A&M University, Kurban Intelligence Lab

02/2025 – Present

Research Assistant with Prof. Hasan Kurban and Prof. Erchin Serpedin
 Topics: All things Perception/Vision from Cog Sci/Phys approach.

Qatar Computing Research Institute, AI Group

05/2025 – 03/2026

Research Intern with Dr. Halima Bensmail and Dr. Michael Aupetit
 Topics: Normalizing Flows and Generative Models.

Texas A&M University, Power System Modeling Lab

01/2024 – 12/2024

Research Assistant with Prof. Selma Awadallah
 Topic: ML for fault detection in electrical transformers.

Industry Experience**Overshoot (Y Combinator W26)**

05/2026 – Present

Research Scientist Intern with Younes El Hjouji and Zakaria El Hjouji
 Topic: Unified Encoder for Vision Language Models.

Ottonomi AI

07/2025 – 10/2025

Computer Vision Software Engineer with Prof. Hussein Alnuweiri
 Under NDA ;)

Digital & Integration, SLB

06/2025 – 08/2025

Engineering Intern with Azhar Khan
 Topic: Automated Well Integrity Evaluation.

Grants and Funding**ValuesLab Student Fellowship Grant, Columbia University**

2026 – 2027

Principal Investigator, Awarded \$3,000 of funding and compute.
 Topic: Machine Imagination as a new learning paradigm.

Student Research Experience Grants (SREG), TAMUQ	2026 – 2027
Lead Undergraduate Investigator, Awarded \$50,000 of funding.	
Topic: Physics informed 3D reconstruction and understanding.	
Undergraduate Research Experience Project (UREP), QRDI	2025 – 2026
Member, Awarded \$26,000 of funding.	
Topic: ML for fault detection in electrical transformers.	
Travel Award to ICVSS in Sicily, Italy, TAMUQ	2026
Travel Award to CVPR in Denver, USA, Overshoot (YC W26) and TAMUQ	2026
Travel Award to ACVSS in Accra, Ghana, Google and INRIA	2026
Travel Award to EACL in Rabat, Morocco, TAMUQ	2026
Travel Award to MenaML in Jeddah, Saudi, KAUST and Google DeepMind	2026
Travel Award to NeurIPS in San Diego, USA, TAMUQ	2025
Travel Award to ICML in Vancouver, Canada, TAMUQ	2025
Travel Award to IFTP in College Station, USA, TAMU	2025
Travel Award for SLEP leadership exchange in College Station, USA, TAMU	2025
Complementary Registration to LOGML in London, England, Imperial College	2026
Complementary Registration NeurIPS, NeurIPS Committee	2025

Selected Talks

Invited Keynote Speaker at the “MusIML” Workshop, NeurIPS 2025	12/2025
Invited Speaker at “Human Inspired Computer Vision” Seminar, TAMUQ	11/2025
Selected Speaker at the “Undergraduate Research Conference”, MIT	10/2025
Invited Speaker at Moroccans in Artificial Intelligence Research (MAIR)	10/2025

Honors and Awards

Richard E. Ewing Award for Excellence in Research (1 out of ~450) – \$1,250	2026
Undergraduate Research Scholars, College of Engineering, TAMU	2026
1st Place at Qatar Computing Research Institute Symposium (1/108)	2025
First Place at AIX Hackathon, Rising Ventures (+85 Teams) – \$5,000	2025
1st Place at EC 3rd Undergraduate Research Retreat – \$550	2025
Woqod x Qatar Foundation Scholar Award Recipient – \$10,000	2025
Second Place (Global Phase) Invent for the Planet by TAMU – \$2,500	2025
First Place (Qatar Phase) Invent for the Planet by TAMUQ – \$1,650	2024
+Best Prototype Award	
+Best Video Award	
Winner of Qatar Foundation Tech Pitch Competition – \$11,000	2024
2nd Place Texas A&M University Qatar Robotics Competition	2024
Inducted in Engineering Honors Program at TAMUQ	2023

Academic Service

Reviewer, NeurIPS, CreativeAI Track	2026
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Reviewer, ICML, AI4Good Workshop	2026
Reviewer, CVPR, ReLearn: Rediscovering Intelligence Workshop	2026
Volunteer, ICML/NeurIPS, MusIML Workshop	2025

Leadership

ACL, <i>Student Member</i>	2026 – Present
IEEE, <i>Student Member</i>	2023 – Present
The Peace Club TAMUQ, <i>Advisor</i>	2025
The Peace Club TAMUQ, <i>President</i>	2025
The Peace Club TAMUQ, <i>Vice-President</i>	2024
Lead Organizer, “Effective Humanitarian Engineering Solutions Workshop”	2024
Qatar Foundation, <i>Student Orientation Leader</i>	2024
IEEE Student Chapter, <i>Class Representative</i>	2023
Moroccan National Swimming Federation, <i>Instructor Volunteer</i>	2022

Languages

English (Fluent) · Arabic (Native) · French (Bilingual Proficiency)
Spanish (Elementary)

References

Hasan Kurban

Assistant Professor, Hamad Bin Khalifa University

Adjunct Associate Professor, Indiana University Bloomington

hkurban@hbku.edu.qa

hakurban@iu.edu

Erchin Serpedin

Department Chair, ECEN Qatar Campus

Full Professor, Texas A&M University

eserpedin@tamu.edu

Halima Bensmail

Principal Scientist, Qatar Computing Research Institute

Previously: Oak Ridge National Lab, University of Washington, INRIA

hbensmail@hbku.edu.qa

Zakaria El Hajouji

Founder & CEO, Overshoot (Y Combinator W26)

Previously: MIT, Meta, Uber

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